

Bill of Materials - t7_r2 HEV cell

VBF-T7R2-BOM-001

Per-cell layer masses (layer_kg_m2 x area 0.1027 m2) and design deltas:

positive electrode NMC811	16.842 g	(75.6 um)
negative electrode graphite	9.859 g	(85.2 um, r 2.0 um, por 0.32)
Al collector	4.437 g	(16 um)
Cu collector	11.042 g	(12 um)
separator polyolefin	0.259 g	(12 um, por 0.47)
total layer mass	42.440 g	thickness 200.8 um

Design deltas vs Chen2020: Negative particle radius 5.86->2.0 um;

Cation transference number 0.2594->0.5; Negative electrode porosity 0.25->0.32;

Total heat transfer coefficient 10->80 W/m2/K (liquid cooling).

Electrolyte mass excluded by contract calc-energy formula (same convention as energy density).